

Algebra

Choose the Appropriate Answer

1. An algebraic expressions which is equivalent to the verbal statement "Three times the sum of x and y" is

(a) $3(x + y)$ (b) $3 + x + y$ (c) $3x + y$ (d) $3 + xy$

Ans: (a) $3(x + y)$

2. The numerical co-efficient of $-7mn$ is

(a) 7 (b) -7 (c) p (d) -p

Ans: (b) -7

3. Choose the pair of like terms

(a) $7p$, $7x$ (b) $7r$, $7x$ (c) $-4x$, 4 (d) $-4x$, $7x$

Ans: (d) $-4x$, $7x$

4. The value of $7a - 4b$ when $a = 3$, $b = 2$ is

(a) 21 (b) 13 (c) 8 (d) 32

Ans: (b) 13

5. The variable in the expression $16x - 7$ is

(a) x (b) 16 (c) - 7 (d) None

Ans: (a) x

6. The constant term of the expression $2y - 6$ is

(a) 2 (b) - 6 (c) y (d) None

Ans: (b) - 6

7. In the expression $25m + 14n$, the type of the terms are _____ terms.

(a) Monomial (b) Polynomial (c) Like terms (d) Unlike terms

Ans: (c) Like terms

8. The number of terms in the expression $3ab + 4c - 9$ is

(a) 2 (b) 4 (c) 1 (d) 3

Ans: (d) 3

9. The numerical co-efficient of the term $-xy$ is

(a) -x (b) - y (c) 1 (d) -1

Ans: (d) -1

10. The addition of $3mn$, $-5mn$, $8mn$ and $-4mn$ is

(a) mn (b) - mn (c) $2mn$ (d) $3mn$

Ans: (c) $2mn$

$3mn + 8mn - 5mn - 4mn = 11mn - 9mn = 2mn$
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11. When we subtract 'a' from '- a', we get

- (a) 0 (b) 2a (c) -2a (d) -a

Ans: (c) -2a

12. In an expression, we can add or subtract only

- (a) Like terms (b) Unlike terms (c) All terms (d) None of the above

Ans: (a) Like terms

13. The addition of -7b and 2b is

- (a) - 5b (b) 9b (c) - 9b (d) 5b

Ans: (a) - 5b

14. The subtraction of 5m from -3m is

- (a) 2m (b) 8m (c) - 8m (d) - 2m

Ans: (c) - 8m

15. The additive inverse of - 37xyz is

- (a) (1 / 37) xyz (b) 37 xyz (c) 73 xyz (d) - 73 xyz

Ans: (b) 37 xyz

16. An expression equated to another expression is called

- (a) An equation (b) Inequation (c) Linear equation (d) None

Ans: (a) An equation

17. If a = 5, the value of 2a + 5 is

- (a) 5 (b) 10 (c) 15 (d) 20

Ans: (c) 15

18. The sum of twice and four times of the variable x is

- (a) 8x (b) 16x (c) 2x (d) 6x

Ans: (d) 6x

19. The generalization of the number pattern 3, 6, 9, 12,... is

- (a) n (b) 2n (c) 3n (d) 4n

Ans: (c) 3n

20. The solution of $3x + 5 = x + 9$ is

- (a) 2 (b) 3 (c) 5 (d) 4

Ans: (a) 2

$$3x + 5 = x + 9 \Rightarrow 3x - x = 9 - 5 \Rightarrow 2x = 4 \Rightarrow x = 2$$

21. The equation $y + 1 = 0$ is true only when y is

- (a) 0 (b) - 1 (c) 1 (d) - 2

Ans: (b) - 1

22. How many terms are there in the expression $7x^3 + 2xy + z - 7y$?

- (a) 3 (b) 4 (c) 5 (d) 6

Ans: (b) 4

23. How many terms are there in the expression $-2p^3 - 3p^2 + 4p + 7$?

- (a) 1 (b) 2 (c) 3 (d) 4

Ans: (d) 4

24. What is the coefficient of x in the expression $4x + 3y$?

- (a) 1 (b) 2 (c) 3 (d) 4

Ans: (d) 4

25. The value of the expression $x^3 + y^3$ when $x = 2$ and $y = -2$ is

- (a) 0 (b) 8 (c) 16 (d) -16

Ans: (a) 0

26. The value of $x^2 - 2x + 1$ when $x = 1$ is

- (a) 1 (b) 2 (c) -2 (d) 0

Ans: (d) 0

27. The result of subtraction of $3x$ from $-4x$ is

- (a) $-7x$ (b) $7x$ (c) x (d) $-x$

Ans: (a) $-7x$

28. The volume of a cube of side $2a$ is

- (a) $4a^2$ (b) $2a$ (c) $8a^3$ (d) 8

Ans: (c) $8a^3$

29. If we add, $7xy + 5yz - 3zx$, $4yz + 9zx - 4y$ and $-3xz + 5x - 2xy$, then the answer is:

- (a) $5xy + 9yz + 3zx + 5x - 4y$ (b) $5xy - 9yz + 3zx - 5x - 4y$
(c) $5xy + 10yz + 3zx + 15x - 4y$ (d) $5xy + 10yz + 3zx + 5x - 6y$

Ans: (a) $5xy + 9yz + 3zx + 5x - 4y$

30. Which of the following is like term as $7xy$?

- (a) 9 (b) $9x$ (c) $9y$ (d) $9xy$

Ans: (d) $9xy$

31. Add $2mn$, $-4mn$, $8mn$, $-6mn$

- (a) 0 (b) $2mn$ (c) $8mn$ (d) $10mn$

Ans: (a) 0

$$2mn + (-4mn) + 8mn + (-6)mn = 0$$

32. Add $4x^2y$, $-3x^2y$, $-7xy^2$, $7xy^2$

- (a) x^2y (b) xy^2 (c) xy (d) $-x^2y$

Ans: (a) x^2y

$$(4 - 3)x^2y + (-7 + 7)xy^2 = x^2y$$

33. Simplify : $p + (p - q) + q + (q - p)$

- (a) p (b) q (c) $p + q$ (d) $p - q$

Ans: (c) $p + q$

34. Simplify : $z^2 + 11z^2 - 5z - 11z^2 + 5z$

- (a) z^2 (b) $-z^2$ (c) $5z$ (d) $-5z$

Ans: (b) $-z^2$

$$(-1 + 11 - 11)z^2 + (5 - 5)z = -z^2$$

35. Subtract $-x y$ from $x y$

- (a) $x y$ (b) $2xy$ (c) $3xy$ (d) $4xy$

Ans: (b) $2xy$

36. Subtract y^2 from $-5y^2$

- (a) $-6y^2$ (b) $6y^2$ (c) y^2 (d) $-5y$

Ans: (a) $-6y^2$

37. Find the value of the expression $x + 2$ for $x = -2$

- (a) 0 (b) 2 (c) -2 (d) 4

Ans: (a) 0

38. Find the value of the expression $100 - 10x^3$ for $x = 0$

- (a) 10 (b) -10 (c) 100 (d) -10

Ans: (c) 100

39. Find the value of the expression $x^2 + 2x + 1$ for $x = -1$

- (a) 0 (b) 1 (c) -1 (d) 2

Ans: (a) 0

40. If the value of the expression $x^2 - 5x + k = 5$ for $x = 0$, then the value of k is

- (a) 2 (b) 3 (c) 4 (d) 5

Ans: (d) 5

$$\text{Given } x = 0 \Rightarrow (0)^2 - 5(0) + k = 5 \Rightarrow k = 5$$

41. Simplify the expression $10 - 3b - 4 - 5b$ and find their values if $x = 3$, $a = -1$, $b = -2$ is

- (a) 22 (b) 33 (c) 45 (d) 5

Ans: (a) 22

$$10-3b-4-5b \Rightarrow (10-4)-(3b+5b) \Rightarrow 6-8b$$

$$\Rightarrow 6-8(-2) \Rightarrow 22.$$

42. When a certain number m , is divided by 5 and added to 8, the result is equal to $3m$ subtracted from 4. The value of m is

- (a) $3/4$ (b) $3/6$ (c) $-5/4$ (d) $5/6$

Ans: (c) $-5/4$

$$4-3m = 8 + m/5 \Rightarrow 20-15m=40+m$$

$$\Rightarrow 20-40=m+15m \Rightarrow -20=16m$$

$$m = -\frac{5}{4}$$

43. The constant term in the expression $1 + x^2 + x$ is

- (a) 1 (b) x (c) x^2 (d) None of these

Ans: (a) 1

44. In a two-digit number, the units digit is x and the tens digit is y . Then the number is

- (a) $10y + x$ (b) $10x + y$ (c) $10y - x$ (d) $10x - y$

Ans: (a) $10y + x$

45. What are the coefficients of x in the expression $8 - x + y$?

- (a) 1 (b) -1 (c) 8 (d) None of these

Ans: (b) -1

46. What are the coefficients of y in the expression $4x - 3y$?

- (a) 4 (b) -3 (c) 3 (d) None of these

Ans: (b) -3

47. What are the coefficients of y in the expression $yz^2 + 5$?

- (a) 5 (b) z (c) z^2 (d) None of these

Ans: (c) z^2

48. Write an expression: Raja's father's age is 5 years more than 3 times Raja's age. If Raja's age is x years, then father's age is

- (a) $3x+5$ (b) $5-3x$ (c) $3x-5$ (d) $15x$

Ans: (a) $3x+5$

49. Identify the coefficient of x in expression $8 - x + y$

- (a) 0 (b) 8 (c) -1 (d) 1

Ans: (c) -1

50. The number of terms in $4p^2q - 3pq^2 + 5$ is

- (a) 7 (b) 3 (c) 1 (d) 4

Ans: (b) 3

51. The expression for sum of numbers a and b subtracted from their product is

- (a) $a + b - a b$ (b) $a b - a + b$ (c) $a b - (a + b)$ (d) $a b + a - b$.
Ans: (c) $a b - (a + b)$

52. The sum of $m n + 5 - 2$ and $m n + 3$ is
 (a) $2mn + 3$ (b) 6 (c) $2mn + 8$ (d) $2mn + 6$.
Ans: (d) $2mn + 6$

53. What is the statement for the expression $3mn + 5$
 (a) Number 5 added to product of number m and n (b) Number 5 added to 3 times the product of m and n.
 (c) 5 more than 3 times the product of the numbers m and n (d) none of these
Ans: (b) Number 5 added to 3 times the product of m and n.

54. The constant term in the expression $1 + x^2 + x$ is
 (a) 1 (b) 2 (c) x (d) x^2
Ans: (a) 1

55. The coefficient of y^3 in the expression $y - y^3 + y^2$ is
 (a) 1 (b) y (c) $-y^2$ (d) -1
Ans: (d) -1

56. The number of terms in the expression $1.2ab - 2.4 b + 3.6a$ is
 (a) 1.2 (b) -2.4 (c) 3.6a (d) 3
Ans: (d) 3

57. What is the numerical coefficient of y^2 in the expression $2x^2y - 15xy^2 + 7$
 (a) $-15 x$ (b) -15 (c) 2 (d) 7
Ans: (a) $-15 x$

58. The expression $x + y - x y$ is
 (a) Monomial (b) Binomial (c) Trinomial (d) Polynomial
Ans: (b) Binomial

59. The expression xyz is
 (a) Monomial (b) Binomial (c) Trinomial (d) Zero polynomial
Ans: (c) Trinomial

60. From the following expressions $10pq, 7p, 8q, -p^2q^2, -7pq, -23, 3a b$.
 The like terms are
 (a) $3, 7p$ (b) $10 p q, -7p q$ (c) $a b, 3a, b$ (d) $10pq, 7p, 8q$
Ans: (b) $10 p q, -7p q$

61. Sum of $3m$ and $2n$ is
 (a) $5mn$ (b) $3m+2n$ (c) $5m$ (d) $5n$
Ans: (b) $3m+2n$

62. Subtract $a - b$ from $a + b$ the result is
 (a) $2a + 2b$ (b) $2a$ (c) $2b$ (d) $2a - 2b$
Ans: (c) $2b$

63. Subtracting $-5y^2$ from y^2 , the result is
 (a) $-4y^2$ (b) $6y^2$ (c) $4 y^2$ (d) $-6 y^2$
Ans: (b) $6y^2$

64. The value of expression $5n - 2$, when $n = -2$ is
(a) -12 (b) 8 (c) 1 (d) -8

Ans: (a) -12

65. The value of expression $7a - 4b$ for $a = 3$, $b = 2$ is
(a) 13 (b) $7a - 6b$ (c) $21a - 8b$ (d) 29

Ans: (a) 13

66. When $x = 0$, $y = -1$, then the value of expression $2x + 2y$ is
(a) 4 (b) 0 (c) -2 (d) 2

Ans: (c) -2

67. Factors of the term $15x^2$ in the expression $15x^2 - 13x$ are
(a) $15, x, x$ (b) $15, -13$ (c) $15x^2, -13x$ (d) 15

Ans: (a) $15, x, x$

68. Factors of the terms $-4p^2q^2$ in the expression $9p^2q^2 - 4p^2q^2$ are
(a) $9p^2q^2, -4p^2q^2$ (b) $9, -4$ (c) $-4, p, q, q$ (d) -4

Ans: (c) $-4, p, q, q$

69. Which of the following is monomial
(a) $2x + 3$ (b) $2x$ (c) $4x + 2y + 3$ (d) $4y + 5x + z - 1$

Ans: (b) $2x$

70. Which of the following is trinomial
(a) $2a + 6b - 1$ (b) 1 (c) $5a - 7$ (d) $a + b + c - 3$

Ans: (a) $2a + 6b - 1$

71. Identify the terms in the expression $x + y + 1$ which are not constant
(a) $x, y, 1$ (b) x, y (c) $x, 1$ (d) y, x, z

Ans: (a) $x, y, 1$

72. Terms with factors y in the expression $8 + xy + xyz$ are
(a) x, y, xyz (b) x, x, z (c) $8, x, y, xyz$ (d) $y, 1$

Ans: (b) x, x, z

73. The value of expression $4x - 3$ at $x = 2$ is
(a) -4 (b) 5 (c) 4 (d) 2

Ans: (b) 5

74. The value of expression $5n^2 + 5n - 2$ for $n = -2$ is
(a) 13 (b) 3 (c) 8 (d) 12

Ans: (c) 8

75. The value of expression $2a^2 + 2b^2 - ab$ for $a = 2$, $b = 1$ is
(a) 2 (b) 8 (c) 6 (d) 10

Ans: (b) 8

76. The value of $x + 7 + 4(x - 5)$ for $x = 2$
(a) -3 (b) 31 (c) 12 (d) 37

Ans: (a) -3

77. The value of expression $2a - 2b - 4 - 5 + a$ at $a = 1$, $b = -2$
(a) 10 (b) -2 (c) 12 (d) -4

Ans: (b) -2

78. What must be subtracted from $2a + b$ to get $2a - b$
(a) $2b$ (b) $4a$ (c) 0 (d) $4a + 4b$

Ans: (a) $2b$

79. What must be added to $3x + y$ to get $2x + 3y$
(a) $5x + 4y$ (b) $-x + 2y$ (c) $x - 2y$ (d) $x + 2y$

Ans: (a) $5x + 4y$

80. Subtract $a + 2b$ from sum of $a - b$ and $2a + b$
(a) $2a - 2b$ (b) $4a + 2b$ (c) $2b$ (d) $-2a + 2b$

Ans: (a) $2a - 2b$

81. On simplifying $(a + b - 3) - (b - a + 3) + (a - b + 3)$ the result is
(a) $a - b + 3$ (b) $a - b - 3$ (c) $3a - b - 3$ (d) $3a + b + c$

Ans: (c) $3a - b - 3$

82. What should be value of 'a' if $y^2 + y - a$ equals to 3 for $y = 1$
(a) -1 (b) -5 (c) 5 (d) 0

Ans: (a) -1

83. What is an expression for the statement: "p is multiplied by 16".

(a) $16p$ (b) $p / 16$ (c) $p + 16$ (d) $p - 16$

Ans: (a) $16p$

84. The expression for the statement: "y multiplied by 10 and then 7 added to the product.

(a) $10 + y + 7$ (b) $7y + 10$ (c) $10y + 7$ (d) $10y$

Ans: (c) $10y + 7$

85. $x + x + x + x =$

(a) 4 (b) $4x$ (c) $3x$ (d) None of these

Ans: (b) $4x$

86. What is the statement for the expression $2y - 9$

(a) $2y$ subtracted from 9 (b) 9 subtracted from y and multiplied by 2
(c) Thrice of y minus 9 (d) 9 subtracted from $2y$

Ans: (a) $2y$ subtracted from 9

87. Give expression for: "5 times of 'y' to which 3 is added".

(a) $y + 15$ (b) $5y + 3$ (c) $(5 / y) + 3$ (d) $3y + 5$

Ans: (b) $5y + 3$

88. If $p = -10$, find the value of $p^2 - 2p - 100$

(a) -20 (b) 120 (c) -220 (d) 20

Ans: (d) 20

89. What should be the value of 'a', if the value of $2x^2 + x - a$ equals to 5, when $x = 0$?

- (a) 5 (b) - 5 (c) 0 (d) 1

Ans: (b) - 5

90. $7n + 20$. To find the 5th term for using the given algebraic expression.

- (a) 45 (b) 55 (c) 35 (d) 65

Ans: (b) 55

91. $2n - 1$. To find the 10th term for using the given algebraic expression.

- (a) 17 (b) 15 (c) 19 (d) 21

Ans: (c) 19

92. $n^2 + 1$. To find the 100th term for using the given algebraic expression.

- (a) 10001 (b) 101 (c) 1001 (d) 11

Ans: (a) 10001

93. Which one of the following is a Monomial?

- (a) $4y$ (b) $3x+2$ (c) $2xz+y$ (d) $5x+6y$

Ans: (a) $4y$

94. Write the numerical coefficient in the expression $-45xyz$.

- (a) 45 (b) - 45 (c) $45xyz$ (d) $-45xyz$

Ans: (b) - 45

95. $x + y + 2x + y + z - 2z =$

- (a) $3x + 2y + z$ (b) $3x - 2y + z$ (c) $3x + 2y - z$ (d) $3x - 2y - z$

Ans: (c) $3x + 2y - z$

96. A trinomial has ----- terms.

- (a) 2 (b) 3 (c) 4 (d) 1

Ans: (b) 3

97. The numerical factor in the term is called -----

- (a) Variable (b) Constant (c) Coefficient (d) None of these

Ans: (c) Coefficient

98. The part of an algebraic expressions are separated by-----

- (a) Variables (b) Operators (c) Constants (d) Coefficients

Ans: (a) Variables

99. A term is a product of -----

- (a) Factors (b) monomials (c) coefficients (d) None

Ans: (a) Factors

100. Algebraic expressions are made up of -----

- (a) Factors (b) Terms (c) Variables (d) None

Ans: (b) Terms